Milestone 2 Submission

Insurance Price Prediction

Submitted By

Group No. 4 [Batch: 2025-26]

Group Members

1. **Anup Kushwaha**
2. **Govind Uttur**
3. **Gunjan Kumar**
4. **Arijit Sen**

Mentor

Mr. Rajeev Kumar

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Milestone 2: Modeling and Business Recommendations

Insurance Price Prediction | Model selection, validation, explainability, and deployment

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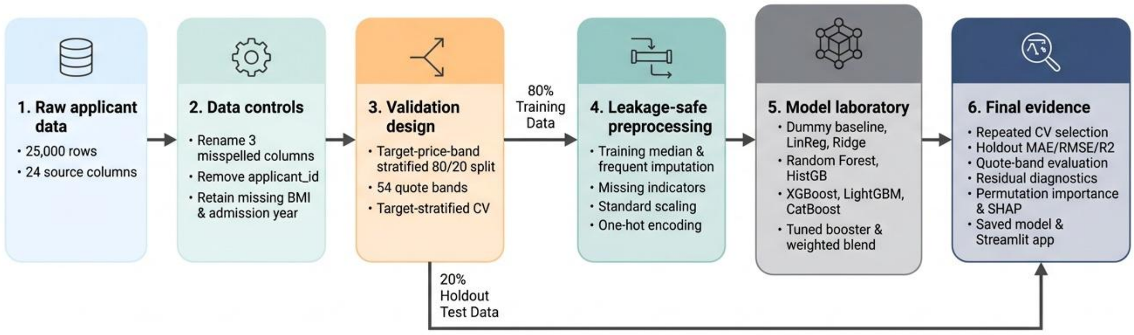
Executive Summary

Milestone 2 converts the Milestone 1 analytical findings into a reproducible insurance-cost modeling workflow. The self-contained notebook starts from Insurance Data.csv, creates a target-price-band-stratified split, trains baseline and advanced models, validates the Milestone 1 engineered-feature proposal, tunes Histogram Gradient Boosting, performs repeated validation, selects the final model, evaluates quote-band post-processing, generates explainability outputs, and saves deployment artifacts.

result	value	meaning
Selected model	50/50 LightGBM and Histogram Gradient Boosting blend	Output label: WeightedBlendLightGBMHistGradient
Test MAE	2,385	Average absolute cost error
Test RMSE	2,972	Large-error-sensitive cost metric
Test R2	0.9568	Variance explained
RMSE improvement vs baseline	79.2%	Relative to mean prediction

Technical interpretation: The selected 50/50 LightGBM and Histogram Gradient Boosting blend (WeightedBlendLightGBMHistGradient) achieved test MAE 2,385, RMSE 2,971.91, and R2 0.9568. Selection was based on repeated target-stratified validation.

Business interpretation: The model reduces holdout RMSE by 79.2% relative to predicting the mean premium, while retaining a controlled decision-support role.



Milestone 2 end-to-end modeling workflow.

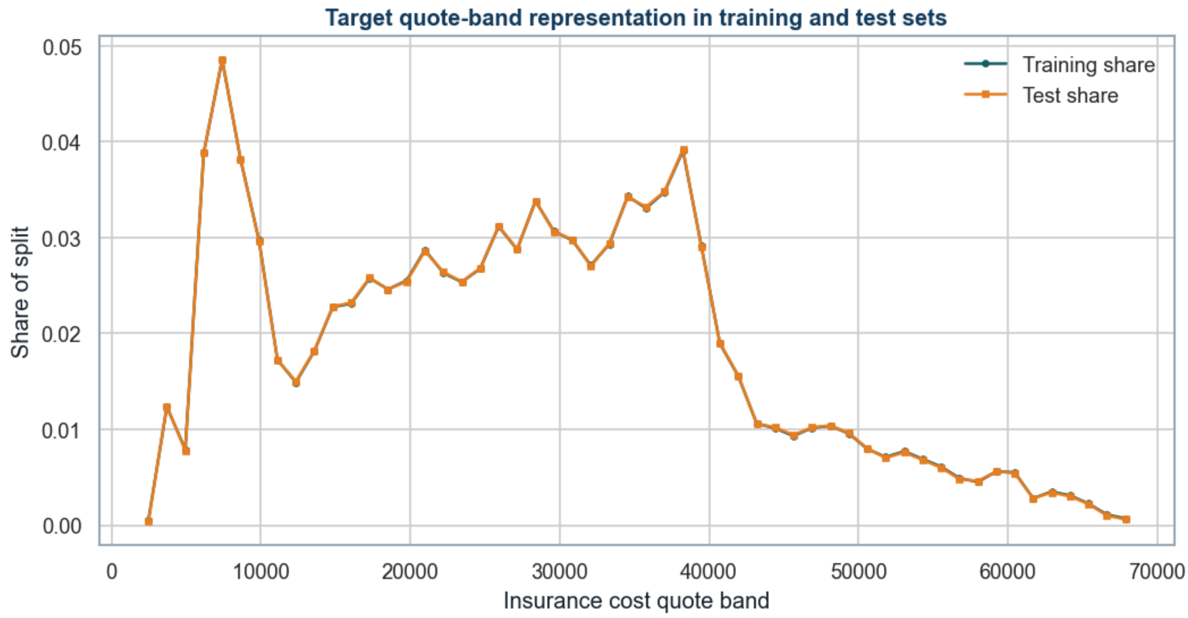
Technical interpretation: The workflow separates data controls, validation, preprocessing, training, selection, and final assessment.

Business interpretation: The diagram shows how raw applicant data becomes a reviewed quote estimate with reproducible evidence.

Introduction

The project predicts insurance_cost from health, lifestyle, demographic, medical-history, and insurance-history variables. The target is numeric but appears on 54 fixed historical quote bands separated by 1,234. Regression remains the primary formulation because business users need a continuous estimate, while quote-band rounding is evaluated as a separate deployment output.

- Input data: 25,000 applicant records and 24 source columns.
- Modeling target: insurance_cost.
- Identifier control: applicant_id is excluded before splitting and training.
- Validation: 80/20 target-band-stratified split with random_state 42.
- Selection: common cross-validation followed by repeated target-stratified validation.
- Deployment: raw continuous estimate, nearest valid quote band, and optional calibrated diagnostic.



Target quote-band representation in the training and test partitions.

Technical interpretation: Both lines overlap closely across all 54 target levels. Each quote band is represented in training and test data.

Business interpretation: The evaluation includes routine and rare premium bands instead of measuring performance only on the most common prices.

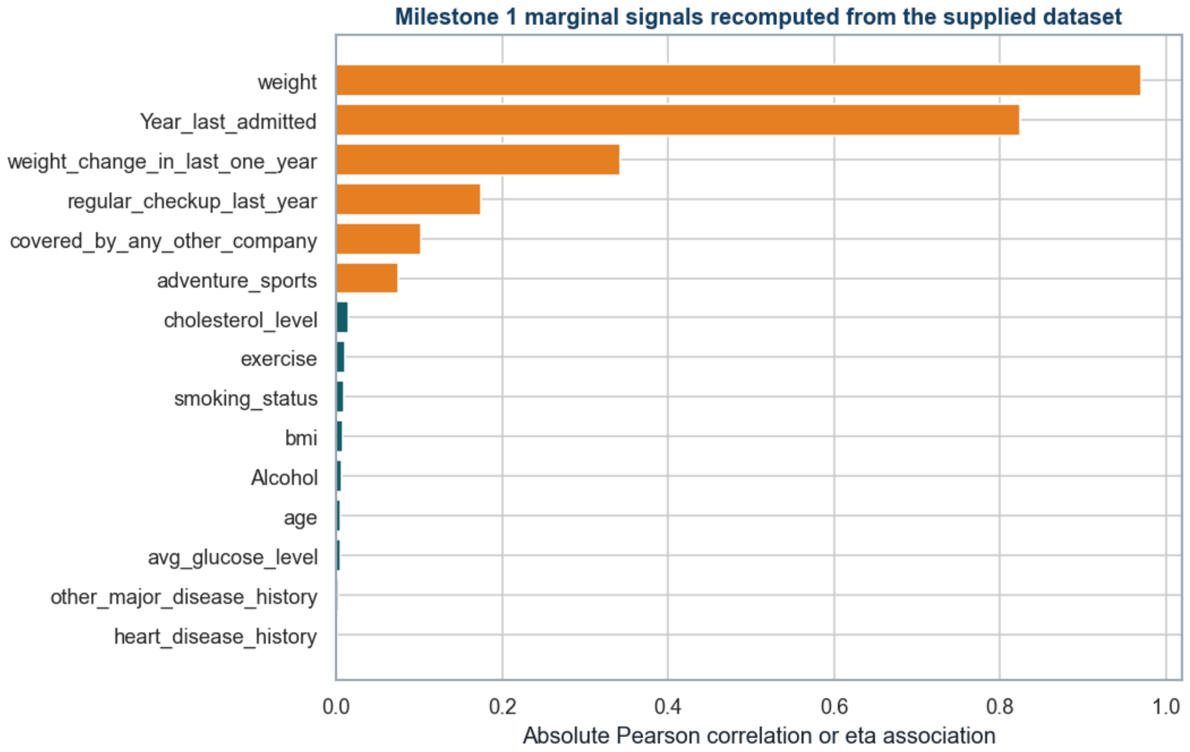
Milestone 1 Handoff

The Milestone 2 notebook does not read Milestone 1 files at runtime. It recomputes the data profile, target-grid structure, missingness, identifier control, target-tail treatment, and marginal signal evidence directly from Insurance Data.csv before model development.

Milestone 1 evidence	Recomputed value	Modeling implication
Dataset rows	25,000.0	Recomputed from the supplied CSV
Source columns	24.000	Includes target and applicant_id
Independent applicant variables	22.000	Target and applicant_id excluded
Valid target quote bands	54.000	Discrete target levels
Target quote-band step	1,234.0	Modal difference between adjacent levels
BMI missing rows	990.0	Retained for pipeline treatment
Year_last_admitted missing rows	11,881.0	Retained as meaningful admission-history missingness
Unique applicant IDs	25,000.0	Identifier excluded before modeling
High-cost rows above IQR fence	0.000	Retained because they remain valid quote-band observations
High-cost rows deleted	0.000	No target-tail deletion

Technical interpretation: The handoff confirms 25,000 rows, 24 source columns, 22 usable applicant variables, 54 target bands separated by 1,234, 990 missing BMI values, 11,881 missing admission-year values, and no deletion of valid high-cost rows.

Business interpretation: Milestone 2 preserves the quote structure and incomplete applicant records while keeping applicant identity outside pricing.



Milestone 1 marginal signal groups recomputed from Insurance Data.csv.

Technical interpretation: Weight, admission year, weight change, regular checkups, other-company coverage, and adventure sports remain the leading marginal signals. The listed lifestyle, medical, age, BMI, glucose, and cholesterol fields remain weak.

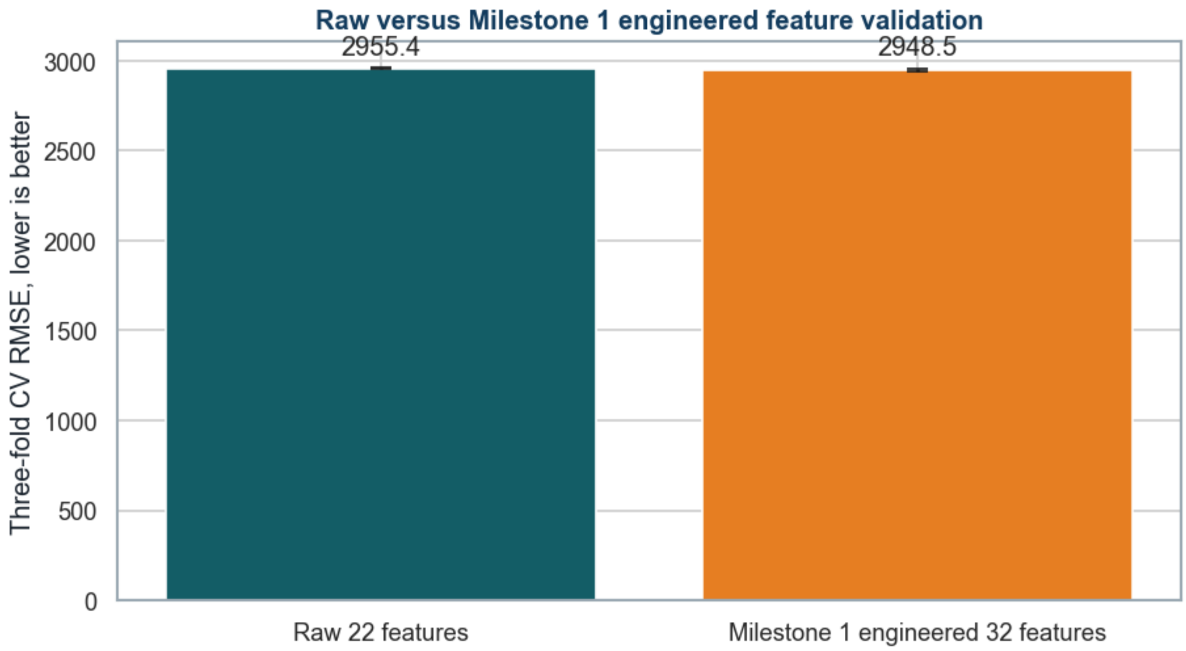
Business interpretation: Strong observed relationships guide model diagnostics, but neither strong nor weak marginal evidence is treated as a causal premium rule.

Milestone 1 Recommendations Implemented

Milestone 1 recommendation	Milestone 2 implementation	Notebook evidence
Use a target-band-aware split	Stratify the holdout and validation folds by the 54 quote bands	Sections 6, 10, and 13
Compare baseline and advanced models	Evaluate mean, linear, tree, boosting, and blend candidates	Sections 9 to 14
Evaluate continuous and quote-band outputs	Compare raw, calibrated, rounded, and calibrated-rounded variants	Section 15
Review residuals by premium segment	Report cost-decile and applicant-segment errors	Section 17
Explain the final model	Use permutation importance and a SHAP challenger explanation	Sections 18 and 19

Raw versus Milestone 1 Engineered Features

feature set	feature count	CV MAE mean	CV RMSE mean	CV RMSE std	selected for downstream modeling
Raw 22 features	22.000	2,368.6	2,955.4	2.752	Yes
Milestone 1 engineered 32 features	32.000	2,359.1	2,948.5	6.745	No



Same-fold validation comparison of raw and Milestone 1 engineered inputs.

Technical interpretation: The engineered set changes mean CV RMSE by 6.92 cost units, a relative gain below the predefined 0.5% requirement, and increases fold-to-fold variability.

Business interpretation: The final model retains the raw 22-feature strategy because the small average gain does not justify less stable validation and additional deployment logic.

Engineered features did not meet the predefined 0.5% RMSE improvement or stability threshold, so the raw feature set is retained for parsimony and portable deployment.

Model Selection and Metrics of Interest

The candidate set progresses from a simple mean baseline to linear, regularized, tree, ensemble, and external boosting methods. This establishes whether additional complexity produces measurable and stable improvement.

Model Reference Guide

The output name is the exact label used in notebook result tables. The reader-friendly explanation states how the model produces a prediction and why it was included.

Baseline and Standard Models

output name	how it works	role and caution
DummyRegressor	Predicts the training-set mean for every applicant.	Defines the minimum performance that a useful model must beat. Caution: Does not use applicant differences.
LinearRegression	Adds a learned contribution from each encoded input to form the prediction.	Provides a transparent additive benchmark. Caution: Cannot naturally represent thresholds and complex interactions.
Ridge	Uses linear regression with an L2 penalty that shrinks unstable coefficients.	Tests whether a more stable linear model improves generalization. Caution: Remains an additive linear model.
DecisionTreeRegressor	Builds if-then split rules that divide applicants into progressively similar groups.	Captures nonlinear thresholds and interactions. Caution: A single tree can overfit and change with the sample.
RandomForestRegressor	Averages many trees trained on resampled rows and randomized feature subsets.	Reduces the instability of one decision tree. Caution: Is larger and less directly interpretable than one tree.
GradientBoostingRegressor	Builds small trees sequentially, with each tree correcting earlier residual errors.	Tests a strong nonlinear boosting benchmark. Caution: Performance depends on learning rate, tree size, and iteration count.
HistGradientBoostingRegressor	Groups numeric values into bins before sequential tree boosting for efficient training.	Provides a fast sklearn boosting candidate for tabular data. Caution: Still requires controls for tree complexity and overfitting.

Advanced and Project-specific Models

output name	how it works	role and caution
RegularizedHistGradientBoosting	The same histogram algorithm with slower learning, leaf limits, and L2 regularization.	Tests a deliberately more constrained configuration. Caution: This is a project label, not a separate software library.
XGBRegressor	An optimized gradient-boosted tree implementation with sampling and regularization controls.	Adds a widely used advanced boosting benchmark. Caution: Requires careful tuning and optional accelerator configuration.
LightGBMRegressor	A histogram-based booster that grows the leaf with the largest expected loss reduction.	Provides efficient learning and strong nonlinear performance. Caution: Leaf-wise growth can overfit when leaf constraints are too loose.
CatBoostRegressor	A boosting implementation designed for stable training and strong categorical-data handling.	Adds a third external booster for comparison. Caution: In this notebook it receives the same one-hot encoded inputs for a fair pipeline comparison.
TunedHistGradientBoostingRegressor	The histogram model whose settings are chosen by RandomizedSearchCV on training folds.	Tests whether systematic tuning improves the base configuration. Caution: Tuning improves the search process but does not guarantee the final winner.
WeightedBlendLightGBMHistGradient	Averages the predictions from LightGBM and regularized Histogram Gradient Boosting.	Tests whether combining two strong but different boosters reduces validation error. Caution: Deployment must load and run both component pipelines.
HistGradientBoostingClassifier	Predicts probabilities for the 54 historical quote-band classes.	Challenges continuous regression with a discrete quote-band formulation. Caution: The multiclass loss does not itself enforce the numeric order of the bands.

Technical interpretation: The comparison covers a baseline, additive linear models, individual trees, bagged trees, sequential boosting, tuned boosting, a weighted blend, and a quote-band classifier.

Business interpretation: Each additional level of complexity must provide stable validation improvement that justifies its operating and governance cost.

Metric Reference Guide

metric	full meaning	interpretation
RMSE	Root mean squared error	Cost-unit error that gives extra weight to large misses
MAE	Mean absolute error	Average absolute error in cost units
R2	Coefficient of determination	Share of target variance explained relative to the mean
MAPE	Mean absolute percentage error	Error as a percentage of actual cost
SMAPE	Symmetric absolute percentage error	Percentage error scaled by actual and predicted cost
Band accuracy	Exact or within-band rate	Share of outputs close to a valid quote band

Technical interpretation: RMSE ranks validation performance; MAE, R2, MAPE, and SMAPE prevent a one-metric conclusion. Band accuracy is assessed after continuous model selection.

Business interpretation: MAE and RMSE communicate monetary error, while quote-band distance connects the model to historical pricing operations.

Model Building and Evaluation

Dependent and Independent Variables

dataset role	column count	columns
Independent variables (X)	22.000	years_of_insurance_with_us, regular_checkup_last_year, adventure_sports, Occupation, visited_doctor_last_1_year, cholesterol_level, daily_avg_steps, age, heart_disease_history, other_major_disease_history, Gender, avg_glucose_level, bmi, smoking_status, Year_last_admitted, Location, weight, covered_by_any_other_company, Alcohol, exercise, weight_change_in_last_one_year, fat_percentage
Dependent variable (y)	1.000	insurance_cost
Excluded before splitting	1.000	applicant_id

Technical interpretation: The model uses 22 independent applicant variables to predict insurance_cost. applicant_id is removed before splitting because it has no generalizable pricing meaning.

Business interpretation: Pricing is based on applicant characteristics and history, not a unique record identifier.

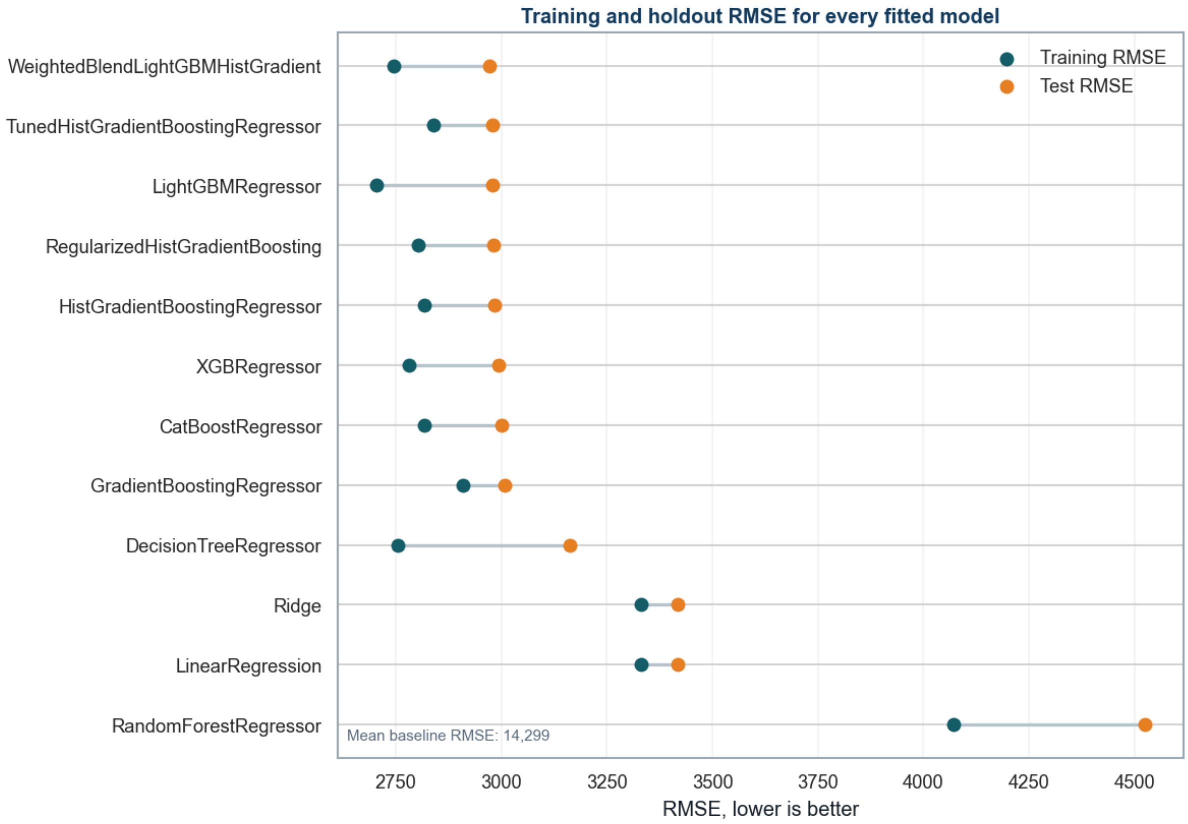
Leakage-safe Modeling Pipeline

field group	action	reason
applicant_id	Removed before split	No generalizable pricing meaning
Numeric fields	Median imputation, missing indicators, scaling	Fitted inside each training fold
Categorical fields	Most-frequent imputation, one-hot encoding	Handles nominal and unseen levels
Target	No preprocessing transform	Keeps errors in cost units

Training and Holdout Results for All Models

model	train MAE	test MAE	train RMSE	test RMSE	train R2	test R2
WeightedBlendLightGBMHistGradient	2,204.6	2,385.2	2,745.3	2,971.9	0.9633	0.9568
TunedHistGradientBoostingRegressor	2,280.9	2,394.4	2,840.1	2,979.1	0.9607	0.9566
LightGBMRegressor	2,170.2	2,391.5	2,704.3	2,979.7	0.9644	0.9566
RegularizedHistGradientBoosting	2,250.2	2,391.3	2,802.3	2,980.1	0.9618	0.9566
HistGradientBoostingRegressor	2,263.0	2,393.6	2,817.8	2,982.5	0.9613	0.9565
XGBRegressor	2,234.8	2,408.4	2,782.1	2,992.8	0.9623	0.9562
CatBoostRegressor	2,264.8	2,416.3	2,818.4	3,001.4	0.9613	0.9559

model	train MAE	test MAE	train RMSE	test RMSE	train R2	test R2
GradientBoostingRegressor	2,339.9	2,421.8	2,909.3	3,007.2	0.9588	0.9558
DecisionTreeRegressor	2,179.5	2,492.4	2,753.8	3,162.4	0.9631	0.9511
Ridge	2,681.8	2,760.4	3,332.0	3,416.7	0.9459	0.9429
LinearRegression	2,682.0	2,760.7	3,332.0	3,416.8	0.9459	0.9429
RandomForestRegressor	3,149.3	3,512.7	4,071.5	4,525.3	0.9193	0.8998
DummyRegressor	11,825.2	11,805.0	14,329.6	14,298.5	0.0000	-0.0000



Training and holdout RMSE comparison for all fitted models; the mean baseline is reported separately on the figure.

Technical interpretation: The table and paired-point chart evaluate every baseline and advanced model on both training and holdout data. The selected blend has the lowest repeated cross-validation RMSE and also retains a controlled train-test gap.

Business interpretation: The final model materially improves pricing accuracy without relying on training performance alone.

Hyperparameter Tuning

technique	implementation	purpose
Randomized hyperparameter search	RandomizedSearchCV over eight reproducible candidates	Covers a broad search space without the cost of a full Cartesian grid
Learning-rate and iteration trade-off	learning_rate with max_iter	Tests slower learning with sufficient boosting iterations

technique	implementation	purpose
Tree-complexity control	max_leaf_nodes and min_samples_leaf	Limits overly specific leaf rules and reduces variance
L2 regularization	l2_regularization	Penalizes overly aggressive fitted responses
Histogram resolution	max_bins	Balances numerical detail with speed and stability
Target-stratified cross-validation	The same three training-only folds for every candidate	Preserves all quote bands and prevents test-set tuning
Overfitting check	Training RMSE compared with cross-validation RMSE	Rejects parameter sets that improve fit without stable validation
Repeated validation and blending	Six repeated folds followed by a 50/50 advanced-model blend	Tests stability and whether complementary model errors can be reduced

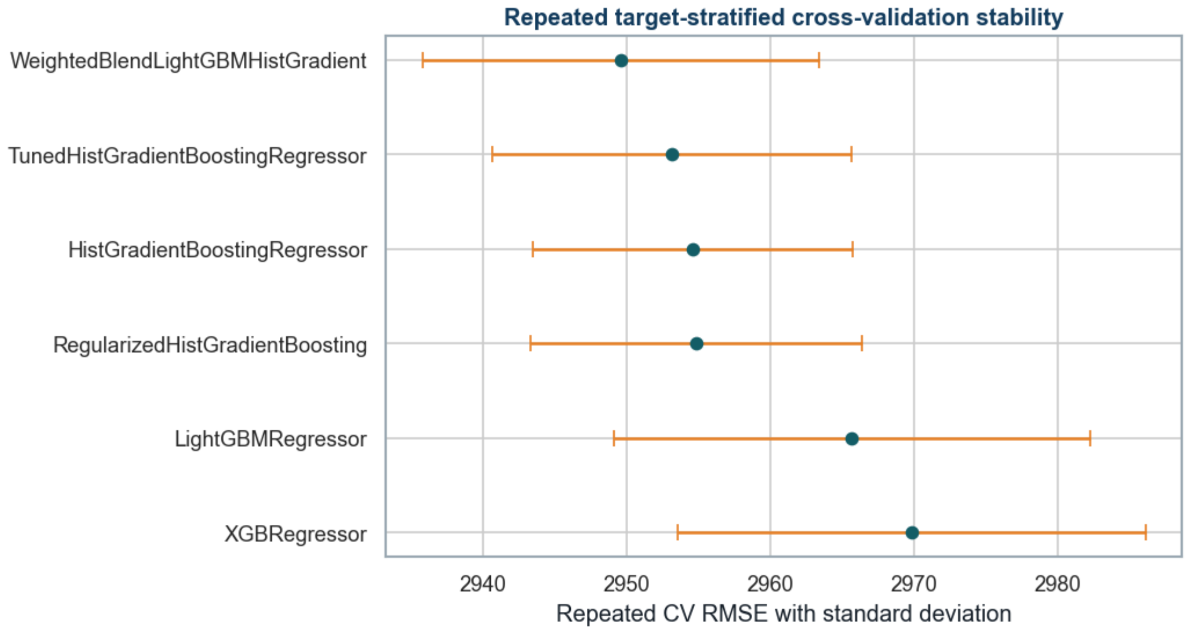
parameter	selected value
model__min_samples_leaf	50.000
model__max_leaf_nodes	23.000
model__max_iter	260.0
model__max_bins	255.0
model__learning_rate	0.030
model__l2_regularization	0.150

Technical interpretation: RandomizedSearchCV tunes iteration count, learning rate, leaf count, minimum leaf support, regularization, and bin count using training-only target-stratified folds.

Business interpretation: The tuning record makes the selected configuration reproducible and reviewable instead of relying on undocumented manual choices.

Repeated Validation

model	repeated cv RMSE mean	repeated cv RMSE std	repeated cv MAE mean	repeated cv R2 mean
WeightedBlendLightGBMHistGradient	2,949.6	13.803	2,363.9	0.9576
TunedHistGradientBoostingRegressor	2,953.2	12.502	2,369.7	0.9575
HistGradientBoostingRegressor	2,954.6	11.138	2,367.0	0.9575
RegularizedHistGradientBoosting	2,954.9	11.554	2,368.6	0.9575
LightGBMRegressor	2,965.7	16.605	2,375.3	0.9572
XGBRegressor	2,969.8	16.297	2,384.7	0.9570

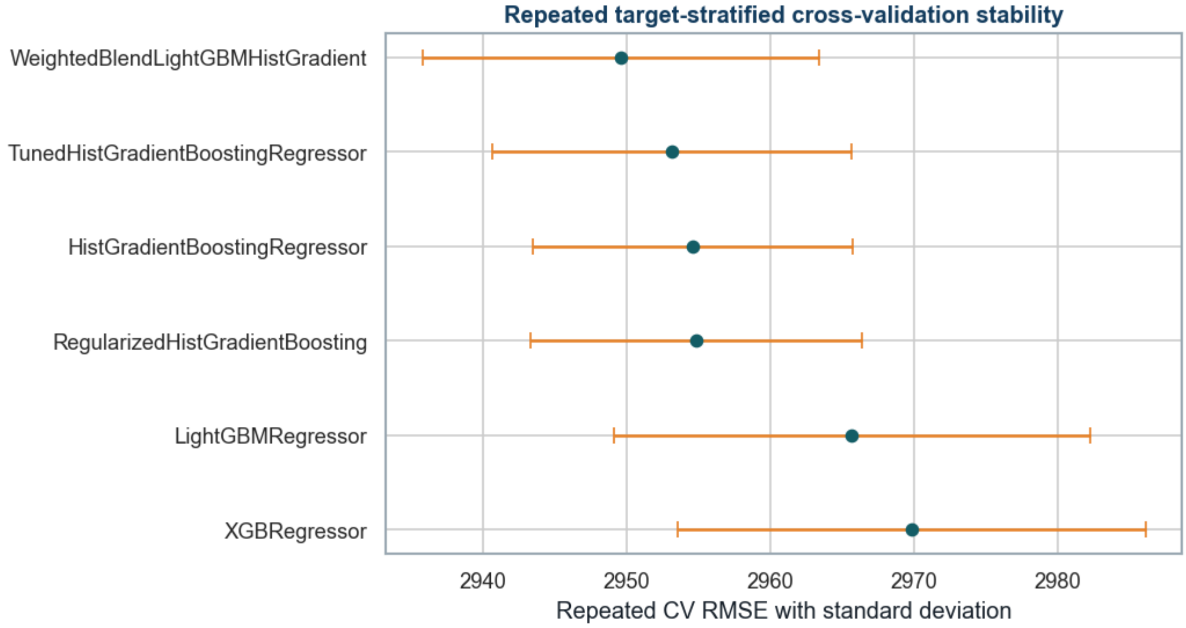


Repeated target-stratified validation mean RMSE and standard deviation.

Technical interpretation: The weighted LightGBM and Histogram Gradient blend has the lowest repeated validation RMSE among the leading candidates.

Business interpretation: The selected model is supported by multiple data partitions, reducing dependence on one favorable split.

Model Comparison and Selection



Repeated target-stratified cross-validation comparison for the leading model candidates.

Technical interpretation: The final model is chosen by repeated training-fold validation. Holdout metrics are reported for rubric comparison, but the selection code does not rank candidates by test performance.

Business interpretation: This explicit rule keeps model selection separate from the reported holdout comparison.

Final Model Decision

Selected Model Name Decoded

name part	meaning	use in final prediction
WeightedBlend	Predictions from more than one fitted model are combined.	An equal 50/50 arithmetic average is used.
LightGBM	Light Gradient Boosting Machine, an efficient histogram-based tree booster.	First component pipeline.
HistGradient	Regularized sklearn Histogram Gradient Boosting regressor.	Second component pipeline.
Complete output	WeightedBlendLightGBMHistGradient	One continuous estimate produced by averaging the two component estimates.

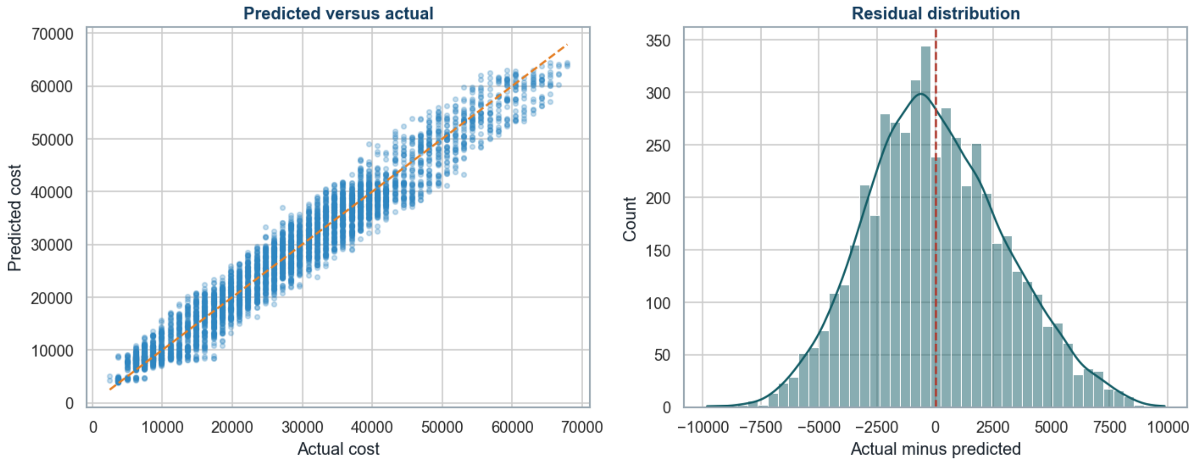
Both components receive the same raw applicant columns through their own leakage-safe preprocessing pipelines. The blend averages two predictions; it does not apply a separate pricing rule.

decision item	result
Selected model	WeightedBlendLightGBMHistGradient
Selection basis	Lowest repeated target-stratified CV RMSE
Repeated CV RMSE	2,949.6

decision item	result
Repeated CV RMSE standard deviation	13.803
Holdout MAE	2,385.2
Holdout RMSE	2,971.9
Holdout R2	0.957

WeightedBlendLightGBMHistGradient achieved the lowest repeated target-stratified CV RMSE (2949.56, standard deviation 13.80) among the leading candidates. Holdout metrics are reported for rubric comparison. The selection code uses only repeated training-fold cross-validation and does not rank candidates by test performance.

Prediction and Residual Diagnostics



Holdout predicted-versus-actual alignment and residual distribution.

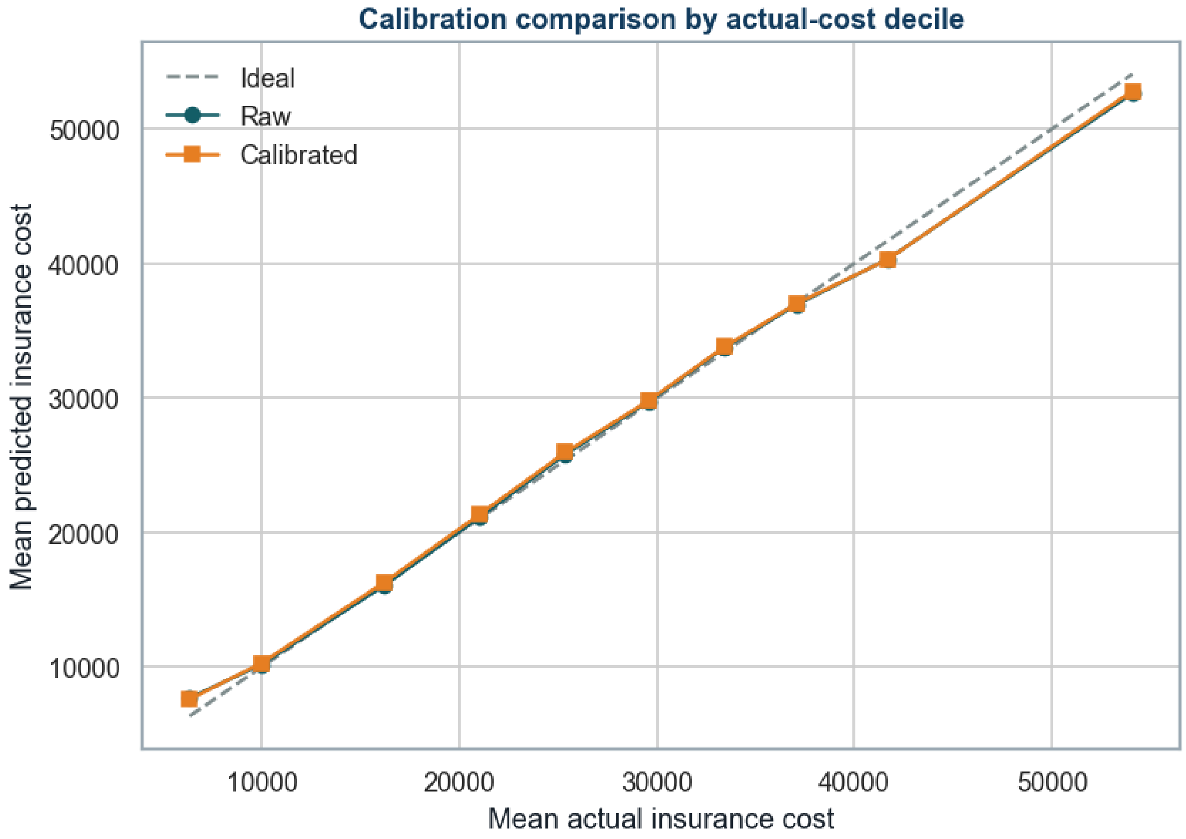
Technical interpretation: Predictions follow the actual-cost line closely. Residuals are centered near zero, with wider errors at some premium levels.

Business interpretation: High-cost or unusual cases need stronger manual review because a small number of large misses drives RMSE above MAE.

Calibration and Quote-band Evaluation

output label	plain-language meaning	role
raw_continuous	The selected regression model's original numeric cost estimate.	Primary analytical estimate
calibrated_continuous	The raw estimate after isotonic calibration.	Secondary diagnostic because MAE and RMSE did not improve
rounded_to_nearest_price_band	The raw estimate moved to the closest one of 54 valid quote bands.	Business quote-band display
calibrated_then_rounded	The calibrated estimate moved to the nearest valid quote band.	Comparison only

variant	MAE	RMSE	R2	MAPE	SMAPE	within 2 band accuracy
raw_continuous	2,385.2	2,971.9	0.9568	11.4579	11.1088	0.6972
calibrated_continuous	2,398.9	2,995.8	0.9561	11.5089	11.1253	0.6916
rounded_to_nearest_price_band	2,368.0	2,998.2	0.9560	11.3183	10.9523	0.6972
calibrated_then_rounded	2,378.2	3,020.7	0.9554	11.3662	10.9323	0.6916



Raw and isotonic-calibrated predictions by actual-cost decile.

Technical interpretation: Calibration does not improve MAE or RMSE in this run. Nearest-band rounding lowers MAE but slightly increases RMSE.

Business interpretation: Use raw prediction for analytical cost estimation, rounded raw prediction for quote-band display, and calibrated cost only as a secondary diagnostic.

Ordinal-style Quote-band Classification Challenger

HistGradientBoostingClassifier predicts probabilities for all 54 historical quote-band classes. The hard-class output chooses the most likely band. The expected-value output multiplies every band cost by its probability and adds the results. The classes correspond to ordered prices, but the multiclass loss does not explicitly enforce their numeric order.

output label	plain-language meaning	role
ordinal_predicted_class	The single quote band with the highest classifier probability.	Hard-class challenger

output label	plain-language meaning	role
ordinal_expected_value	Probability-weighted average of all quote-band costs.	Smoother classifier-based cost challenger

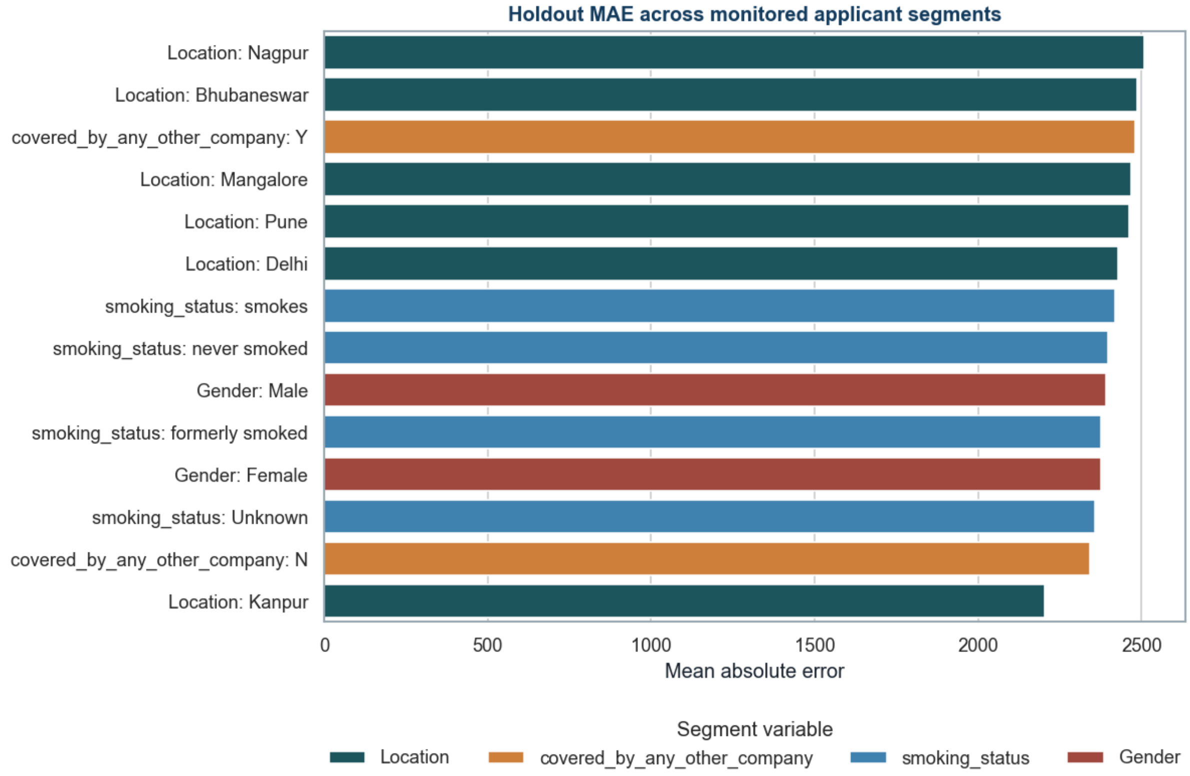
variant	MAE	RMSE	R2	exact band accuracy	within 2 band accuracy
ordinal_predicted_class	2,782.9	3,681.0	0.9337	0.1782	0.6298
ordinal_expected_value	2,442.1	3,068.0	0.9540	0.1656	0.6848

Technical interpretation: The 54-class Histogram Gradient Boosting challenger is weaker than the selected regression model on monetary metrics. Its probability-weighted expected value performs better than hard class prediction but remains behind regression.

Business interpretation: The quote grid should remain a display and monitoring layer rather than replacing the stronger continuous regression model.

Segment-level Error Review

segment variable	segment value	records	MAE	RMSE	mean residual
Location	Nagpur	351.0	2,507.7	3,078.9	33.931
Location	Surat	289.0	2,500.7	3,144.7	-45.594
Location	Bhubaneswar	362.0	2,487.4	3,107.7	155.2
covered_by_any_other_company	Y	1,566.0	2,480.9	3,094.7	156.5
Location	Mangalore	359.0	2,467.4	3,124.0	276.8
Location	Pune	347.0	2,462.7	2,998.7	191.3
Location	Guwahati	338.0	2,444.3	3,019.6	-65.548
Location	Delhi	359.0	2,426.7	3,080.2	-128.2
smoking_status	smokes	788.0	2,418.3	3,027.7	-13.163
smoking_status	never smoked	1,809.0	2,398.1	2,983.5	61.395
Location	Jaipur	346.0	2,395.7	2,955.3	213.7
Gender	Male	3,246.0	2,390.9	2,976.1	-9.228



Holdout MAE for monitored applicant segments.

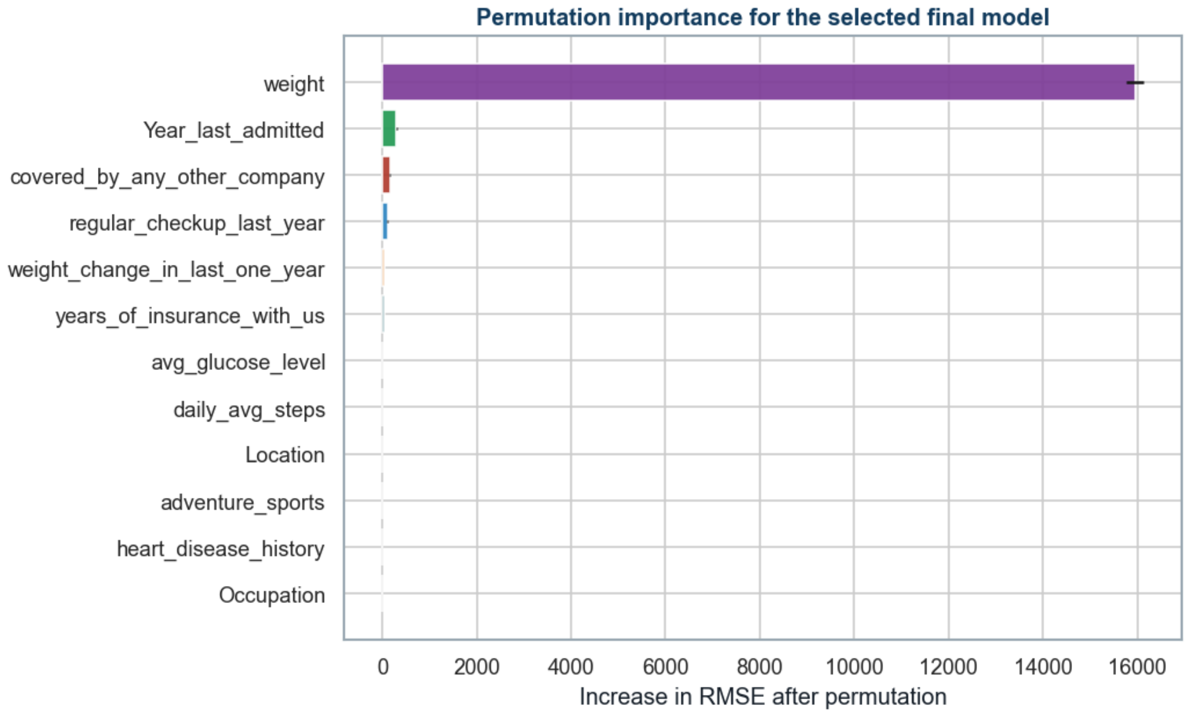
Technical interpretation: Segment errors are similar in magnitude but not identical. The table also reports mean residual to expose systematic overprediction or underprediction.

Business interpretation: Gender, location, coverage, and smoking variables require policy, fairness, and proxy-risk review before production use.

Model Explainability

feature	importance mean	importance std
weight	15,951.9	180.4
Year_last_admitted	301.5	17.380
covered_by_any_other_company	158.4	18.427
regular_checkup_last_year	112.6	13.681
weight_change_in_last_one_year	51.345	5.781
years_of_insurance_with_us	43.173	5.094
avg_glucose_level	6.250	1.322
daily_avg_steps	2.533	3.275
Location	1.147	2.150
adventure_sports	0.651	0.821
heart_disease_history	0.410	0.737

feature	importance mean	importance std
Occupation	0.214	0.450



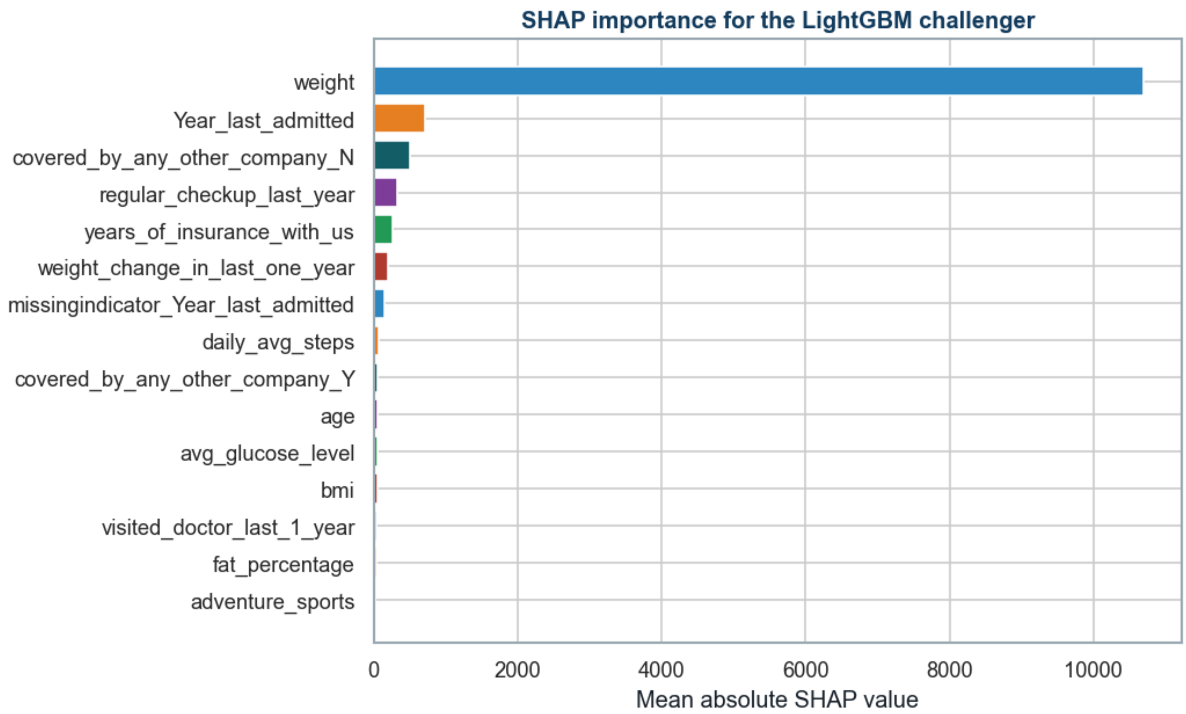
Permutation importance for the selected final pipeline.

Technical interpretation: Weight is the dominant raw input, followed by admission year, other-company coverage, regular checkups, and weight change.

Business interpretation: The strongest drivers should be checked against approved pricing policy and monitored for drift. Importance is association, not causation.

feature	mean abs shap
weight	10,688.3
Year_last_admitted	716.8
covered_by_any_other_company_N	503.8
regular_checkup_last_year	334.2
years_of_insurance_with_us	255.9
weight_change_in_last_one_year	197.4
missingindicator_Year_last_admitted	144.8
daily_avg_steps	61.302
covered_by_any_other_company_Y	56.497
age	53.538

feature	mean abs shap
avg_glucose_level	52.575
bmi	46.559



SHAP importance for the LightGBM challenger.

Technical interpretation: SHAP confirms the leading role of weight and admission history and adds transformed-category detail for the LightGBM model.

Business interpretation: SHAP supports portfolio and case explanation but does not establish that changing an input will cause a premium change.

Business Insights and Recommendations

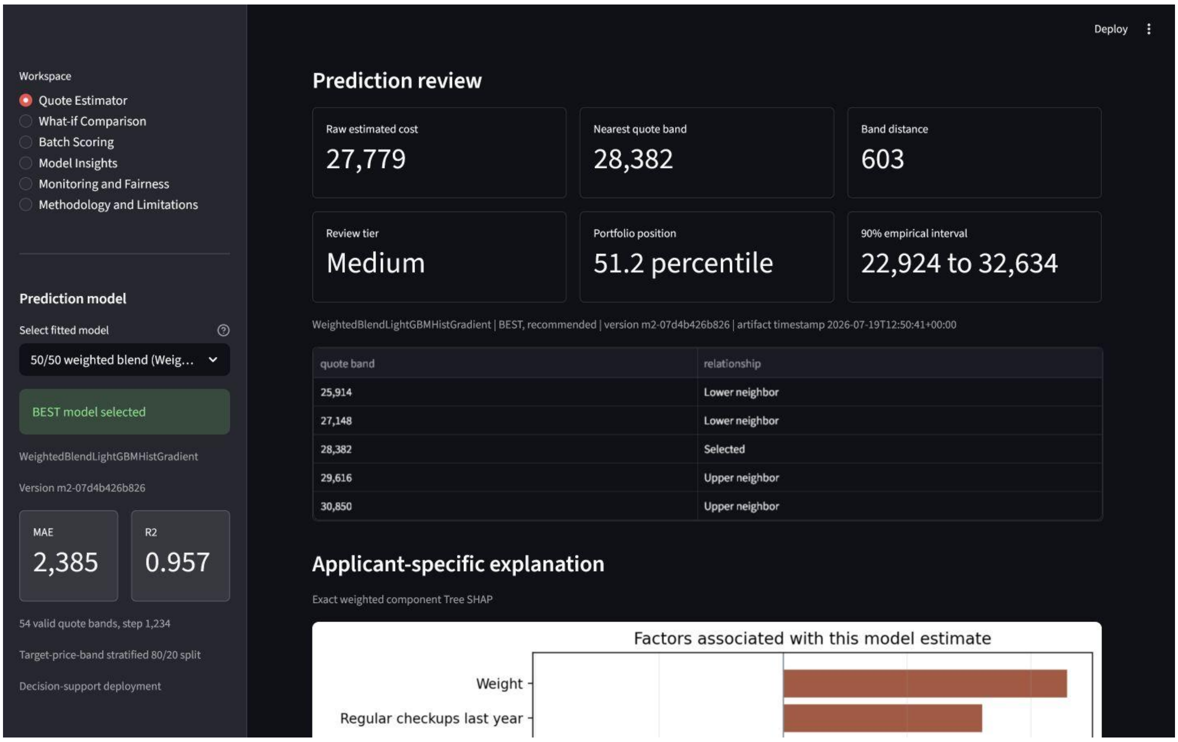
Streamlit Deployment

Run the local application from the project root with: streamlit run app.py

The operational interface contains six workspaces: Quote Estimator, What-if Comparison, Batch Scoring, Model Insights, Monitoring and Fairness, and Methodology and Limitations.

The app allows comparison across 13 fitted regression models. WeightedBlendLightGBMHistGradient is labelled BEST and recommended because it has the lowest repeated target-stratified training-fold CV RMSE among the leading candidates. Choosing a challenger does not change the documented best-model decision.

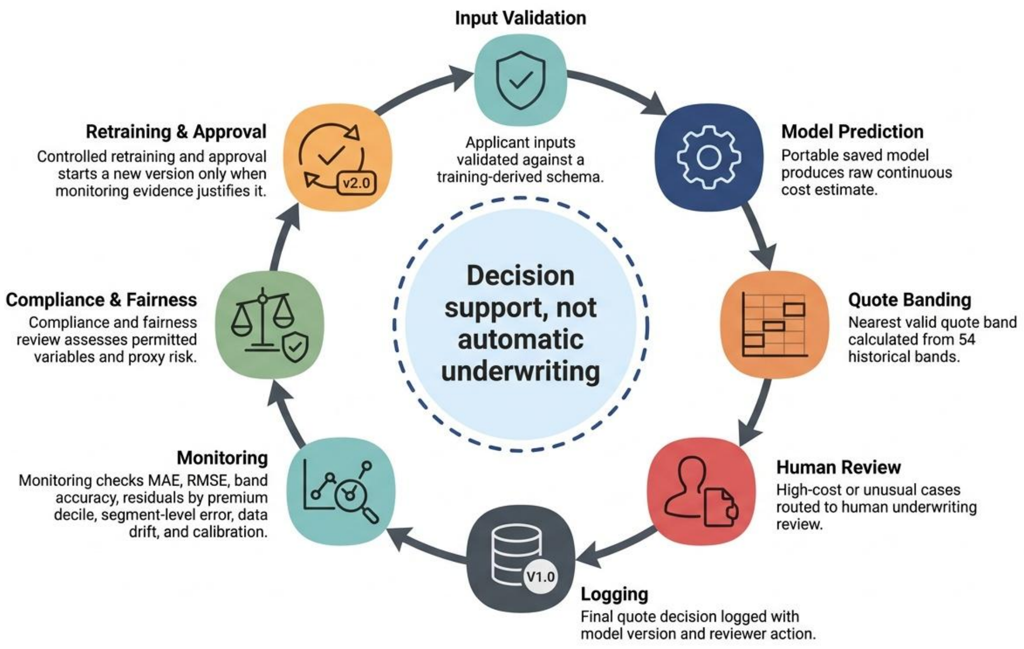
- A synchronized desktop and mobile model selector lists every fitted regression model and keeps the best model clearly identified.
- Quote Estimator reports the raw estimate, nearest valid quote band, band distance, and review tier for any selected model. Out-of-fold portfolio percentile and empirical residual intervals are shown only for the recommended final blend.
- Applicant explanations use weighted component Tree SHAP for the complete fitted blend when the additivity check passes. Challenger models use a clearly labelled model-agnostic one-feature perturbation explanation.
- What-if Comparison stores a baseline, compares a second submitted scenario, and reports sequential input-change associations without presenting them as causal premium advice.
- Batch Scoring validates required columns, types, missing values, categories, and training ranges; invalid rows remain reviewable while valid rows are scored and exportable.
- Monitoring and Fairness reports holdout segment error, prediction distributions, quote-band distributions, and session-only PSI drift against a training reference profile.
- CSV and in-memory PDF exports retain an optional case reference but never send applicant_id into the fitted model.



Streamlit quote review with the best model clearly labelled and alternative fitted models available for comparison.

Technical interpretation: The tested record shows the continuous estimate, nearest valid band, distance to that band, review tier, portfolio position, empirical interval, neighboring bands, and applicant-specific model contributions.

Business interpretation: The interface is decision support. Contributions are associations, uploaded batch monitoring is an offline demonstration, and final underwriting requires documented human, compliance, fairness, and security review.



Responsible deployment and monitoring workflow.

Technical interpretation: The workflow separates prediction, quote-band display, human review, monitoring, compliance review, and controlled retraining.

Business interpretation: The model remains decision support, with a documented reviewer and model version attached to the final business action.

recommendation	evidence	action	measure
Two-value quote display	Target follows 54 fixed quote bands	Show raw estimate and nearest valid band	MAE, RMSE, exact and within-two-band accuracy
High-cost manual review	Large misses drive RMSE above MAE	Review highest predicted-cost and unusual cases	Review rate, corrected error, turnaround time
Driver and drift monitoring	Weight and admission history dominate importance	Compare input distributions and importance by release	Drift thresholds, segment error, calibration
Fairness and policy control	Segment errors vary and proxy risk exists	Review permitted variables with compliance specialists	Segment MAE, RMSE, residual and outcome gaps
Controlled model lifecycle	Notebook produces versioned artifacts and QA	Version data, model, schema, thresholds and approvals	Validation status, incidents, retraining decisions

- Use the model as pricing decision support, not automatic underwriting.
- Route extreme predicted premiums and unusual input combinations for manual review.
- Keep calibration secondary because it does not improve MAE or RMSE in this run.
- Review gender and location use with legal, compliance, and underwriting specialists.
- Revalidate after data drift, policy changes, model changes, or material segment-error movement.

Technical interpretation: The recommendations link model evidence to a named action and monitoring measure. They address accuracy, quote-band operations, explainability, fairness, and lifecycle control.

Business interpretation: The expected benefit is more consistent pricing review without removing human accountability or policy oversight.

Appendix

Reader Glossary

The following terms appear in the notebook, result tables, figures, or deployment recommendations.

term	plain-language meaning	use in this project
Supervised regression	Learning a numeric outcome from examples that contain both inputs and the known outcome.	The primary task predicts continuous insurance_cost.
Target quote band	One of the 54 discrete historical insurance_cost levels.	Used for stratification, rounding, and band-accuracy evaluation.
Stratification	Splitting data while preserving the relative frequency of defined groups.	Keeps all target quote bands represented in training, validation, and test partitions.
Holdout test set	Data kept outside fitting and tuning; its metrics are reported after fitting but are not used by the selection code.	The 20 percent test partition estimates final generalization performance.
Cross-validation (CV)	Repeatedly fitting on part of the training data and validating on the remaining fold.	Provides comparable training-only evidence for model ranking.
Repeated cross-validation	Running cross-validation across multiple shuffled fold assignments.	Checks whether the leading model remains accurate across partitions.
Standard deviation	The amount by which fold results vary around their mean.	A smaller validation value indicates more stable performance.
Data leakage	Information from validation or test data influencing model fitting.	Prevented by fitting preprocessing inside each training fold.
Pipeline	One saved object that applies preprocessing and then the model in a fixed order.	Ensures training and deployment use the same transformations.
Imputation	Replacing missing values using a rule learned from training data.	Uses medians for numeric inputs and the most frequent value for categories.
One-hot encoding	Converting each category into separate zero-or-one indicator columns.	Makes nominal fields usable by all compared estimators.
Standard scaling	Centering numeric inputs and scaling them by their training-set spread.	Supports stable linear-model fitting within the common pipeline.
Hyperparameter	A model setting chosen before fitting, such as learning rate or leaf count.	RandomizedSearchCV tests candidate Histogram Gradient Boosting settings.
Residual	Actual cost minus predicted cost.	Positive values indicate underprediction and negative values indicate overprediction.
Isotonic calibration	A monotonic post-model mapping learned from predictions and observed outcomes.	Tested as a secondary correction for systematic prediction bias.
Nearest-band rounding	Moving a continuous prediction to the closest valid historical quote band.	Produces the business-facing quote-band display.
Permutation importance	The increase in model error after one input column is randomly shuffled.	Explains the selected final pipeline at the raw-input level.
SHAP	Shapley Additive Explanations, an attribution method that assigns feature contributions to predictions.	Explains the LightGBM challenger after preprocessing.
Model drift	A change in input patterns or prediction relationships after deployment.	Triggers monitoring, review, and possible controlled retraining.

term	plain-language meaning	use in this project
Smoke test	A small execution check that confirms a saved model loads and returns finite predictions.	Verifies the portable final_model.pkl artifact.

Rubric Verification

rubric block	report evidence	status	expected score
Model Selection and Metrics	Baseline and advanced families; MAE, RMSE, R2, MAPE, SMAPE, band metrics	Covered	10/10
Model Building and Evaluation	Explicit X/y split, leakage-safe pipeline, baseline and advanced train/test metrics, tuning	Covered	10/10
Model Comparison and Selection	Full train/test comparison, repeated CV, final reason, diagnostics, explainability	Covered	10/10
Business Insights and Recommendations	Evidence, action, measure, governance, monitoring, further steps	Covered	10/10

Rubric verification result: all four 10-mark blocks are explicitly covered. Expected score based on content coverage is 40/40, subject to evaluator judgment.

Reproducibility

- Place Insurance Data.csv in the project root.
- Open notebooks/02_milestone2_modeling.ipynb.
- Run all cells from top to bottom.
- The notebook creates model, table, figure, schema, app, and QA artifacts.
- The notebook imports no local project modeling script.
- Clean-room QA status: passed. Only 02_milestone2_modeling.ipynb and Insurance Data.csv were present before execution.
- The clean-room notebook produced zero error, warning, or stderr output blocks.
- The saved final model loads in a fresh Python process without a custom transformer.

Key Saved Artifacts

artifact	path
Notebook	notebooks/02_milestone2_modeling.ipynb
Notebook HTML	notebooks/02_milestone2_modeling.html
Milestone 1 handoff	outputs/tables/milestone1_handoff_summary.csv
Feature-set comparison	outputs/tables/milestone1_feature_set_comparison.csv
Final model	outputs/models/final_model.pkl
Model metrics	outputs/models/model_metrics.csv
Train/test evaluation	outputs/models/train_test_model_evaluation.csv
Final summary	outputs/models/final_model_summary.csv
Repeated CV	outputs/models/repeated_cv_metrics.csv

artifact	path
Quote-band evaluation	outputs/models/price_grid_evaluation.csv
Feature importance	outputs/models/feature_importance.csv
SHAP importance	outputs/models/shap_importance.csv
Deployment schema	outputs/models/app_schema.json
OOF interval metadata	outputs/models/prediction_interval_metadata.json
Portfolio reference	outputs/models/portfolio_prediction_reference.csv
Training drift profile	outputs/models/training_reference_profile.json
Local explanation reference	outputs/models/local_explanation_reference.json
App fairness summary	outputs/models/app_fairness_summary.csv
App version metadata	outputs/models/app_version_metadata.json
Streamlit app	app.py
QA summary	outputs/reports/milestone2_qa_summary.json

Final Conclusion

The selected 50/50 LightGBM and Histogram Gradient Boosting blend (WeightedBlendLightGBMHistGradient) is supported by the lowest repeated target-stratified CV RMSE and achieves holdout MAE 2,385, RMSE 2,971.91, and R2 0.9568. The continuous regression estimate remains the analytical output. The raw 22-feature strategy is retained because the Milestone 1 engineered challenger did not meet the predefined practical accuracy or stability rule. Nearest-band rounding remains the quote display, and calibration remains a secondary diagnostic. Production use requires underwriting validation, compliance and fairness review, monitoring, security testing, and documented human approval.